

Crop Yield ETL Pipeline

Data-driven agriculture with Apache Spark

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Introduction



Purpose of Pipeline

Enable precise agricultural insights using data



Pipeline Components

Raw ingestion, transformation, storage, visualization



Role of Apache Spark

Fast, scalable distributed data processing engine

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Problem Statement & Dataset

Challenge

Efficiently process and analyze diverse agricultural data streams for accurate crop yield prediction.

Data complexity and volume require scalable, real-time processing.

Dataset includes

- Weather patterns
- Soil metrics
- Rainfall data

Agricultural Influencing Factors

Soil fertility, rainfall, temperature, and weather conditions significantly impact crop productivity.

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Technology Used

Apache Spark

Distributed compute engine for fast ETL

Scala

Efficient functional programming language

Parquet

Columnar storage format for efficient querying

XChart

Visualization library for charts

PostgreSQL

Reliable relational data storage

IntelliJ

IDE for development efficiency

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Architecture Diagram

Raw Layer

CSV files ingestion

Silver Layer

Transform & feature engineering



Bronze Layer

Parquet storage

Gold & Delivery

Final dataset to Postgres & Power BI

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Bronze Layer: Raw Data Ingestion

Transform raw CSV files into Parquet format for optimized storage and processing

```
val bronzeDF = spark.read.csv("raw_data.csv")bronzeDF.write.parquet("bronze_data.parquet")
```

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Silver Layer: Data Transformation

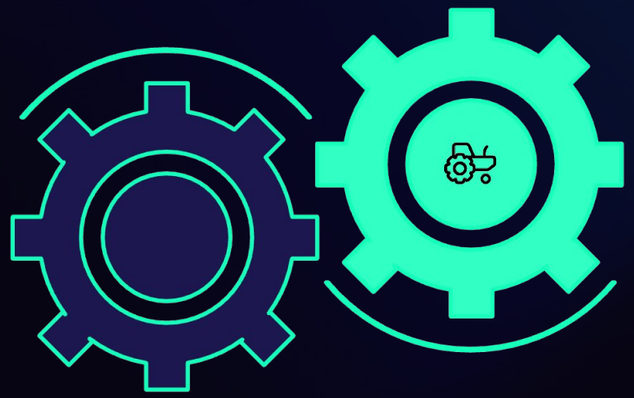
Feature Engineering

Rainfall deviation

Support Level

One Hot Encoding

Climate Stress Index



Normalize Data

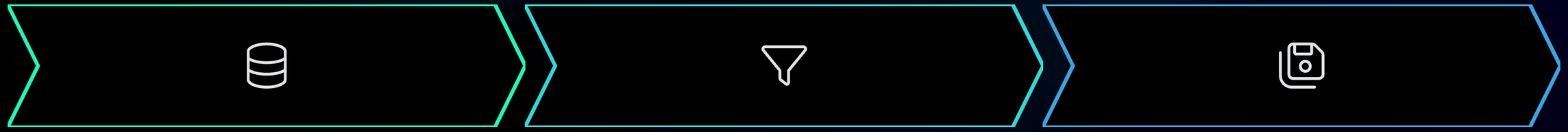
rainfall(mm-cm)

Round temp values

cast boolean columns

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Gold Layer: Final Data Assembly



Load Silver Parquet

Read Silver Layer parquet file

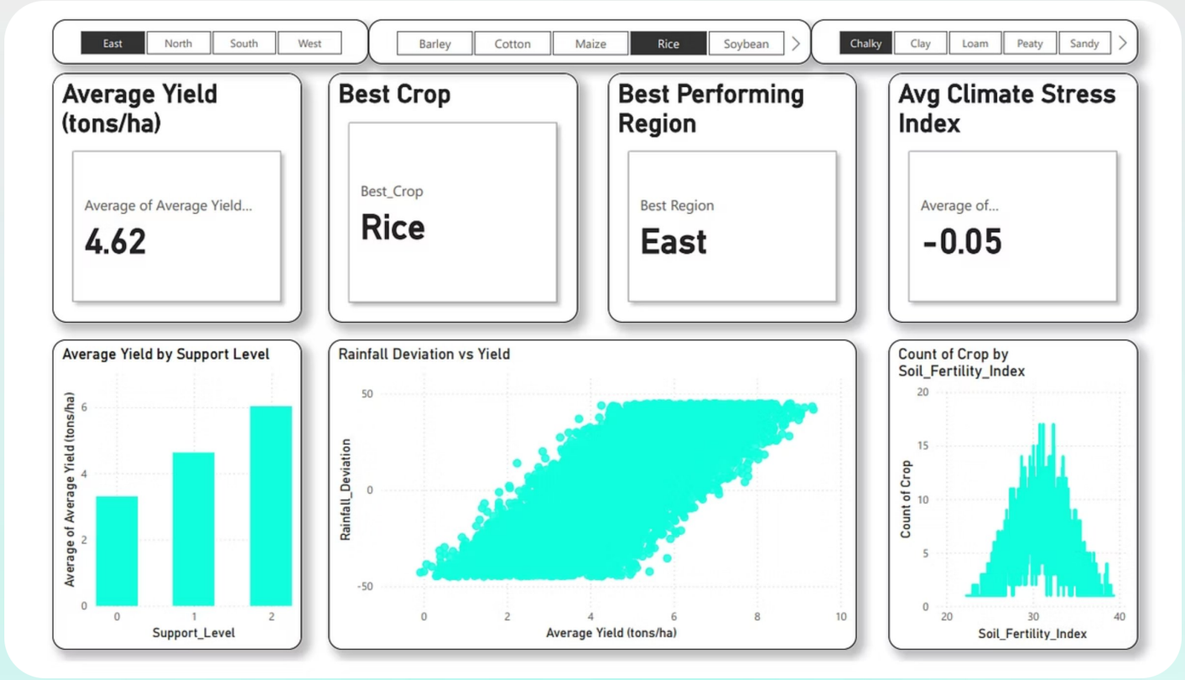
Select Relevant Columns

Choose important fields for analysis

Store Final Data

Save as Parquet, CSV, and PostgreSQL

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Power BI Dashboard Preview

Interactive insights driving smarter farming decisions

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Challenges

Setup

Configuring Spark with Hadoop on Windows required setting up `winutils.exe` to enable saving Parquet files correctly.

Standardization

Rainfall data was in millimeters, temperature in Fahrenheit — needed conversion and normalization for consistent analysis.

Visualization

Spark DataFrames are incompatible with Python visualization libraries. Conversion to Pandas was needed, limiting scalability.

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Key Features of Our Project

End-to-End Workflow

Complete Spark-based ETL pipeline using Bronze, Silver, and Gold layers for structured data lake architecture.

Feature Engineering

Domain-specific features like rainfall deviation, soil fertility index, and climate stress index enrich the dataset.

Visualization

Visual charts generated to compare key agricultural parameters and trends for insightful analysis.

Storage

Data saved both locally and in PostgreSQL for reliable and flexible access.

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